

Inverse Problems in Geophysics

Part 8: Non-linear inversion

2. MGPY+MGIN

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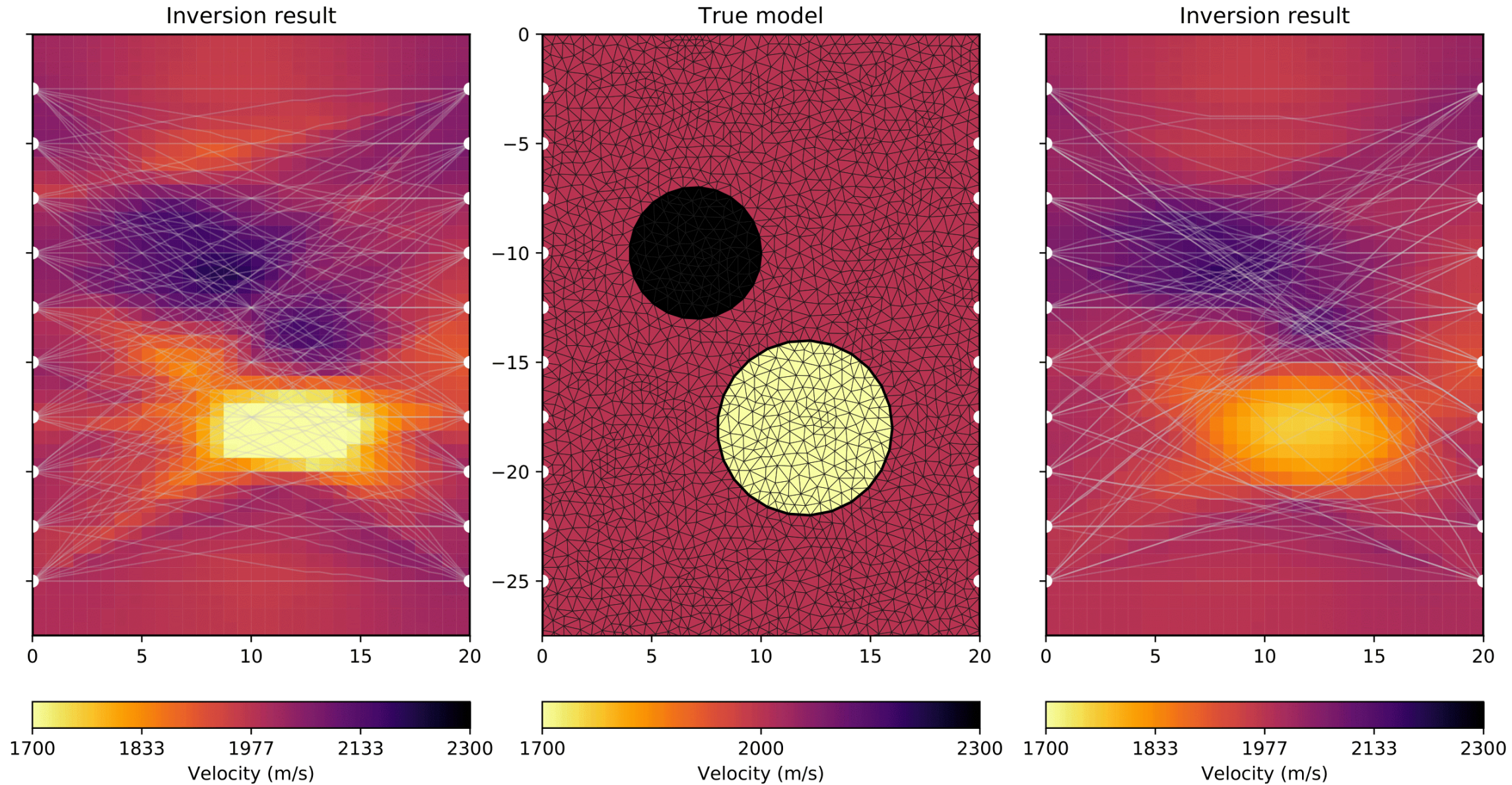
Non-linear inversion

Up to now: $\mathbf{f}(\mathbf{m}) = \mathbf{G}\mathbf{m}$ (with constant $\mathbf{G}$)

Now: either there is no $\mathbf{G}$ or $\mathbf{G}$ depends on $\mathbf{m}$

Example: traveltimes tomography with curved rays

Comparison straight vs. curved rays



Linearization of non-linears

Non-linear problem $y = y_0 e^{\beta x}$ with $\mathbf{m} = (y_0, \beta)^T$

Data transformation: $\hat{y} = \log y = \log y_0 + \beta x$

$\Rightarrow$ Application of linear inverse theory

Logarithmic model transform (positivity constraint)

$\hat{m}_i = \log m_i \Rightarrow$ non-linear problem for $\hat{\mathbf{m}}$ even if linear for $\mathbf{m}$

Linearization using Taylor series

One-dimensional function $f(x)$ of single parameter x :

$$f(x) = f(x_0) + f'(x_0) \cdot (x - x_0) + f''(x_0)/2 \cdot (x - x_0)^2 + \dots$$

Multi-dimensional function of model parameters $m_1, m_2, \dots$:

$$f(\mathbf{m}) = f(\mathbf{m}_0 + \Delta\mathbf{m}) \approx f(\mathbf{m}_0) + \sum_i^M \frac{\partial f(\mathbf{m}_0)}{\partial m_j} \Delta m_j$$

Linearization

$$\mathbf{f}(\mathbf{m}_0 + \Delta\mathbf{m}) = \mathbf{f}(\mathbf{m}_0) + \sum_i^M \frac{\partial \mathbf{f}(\mathbf{m}_0)}{\partial m_j} \Delta m_j = \mathbf{f}(\mathbf{m}_0) + \mathbf{S} \Delta\mathbf{m}$$

We hope the change in the model fits our data: $\mathbf{f}(\mathbf{m}_0 + \Delta\mathbf{m}) \approx \mathbf{d}$

This leads to a linearized problem

$$\mathbf{S} \Delta\mathbf{m} = \mathbf{d} - \mathbf{f}(\mathbf{m}_0) = \Delta\mathbf{d}$$

$\mathbf{S}(\mathbf{m}_0)$: sensitivity matrix of partial derivatives $S_{ij} = \frac{\partial f_i(\mathbf{m}_0)}{\partial m_j}$

Computation of the sensitivity (Jacobian)

- analytically (derivation of the forward operator)
- transforming the PDE and its numerical solution
- perturbation method (brute force)

$$\frac{\partial f_i(\mathbf{m})}{\partial m_j} \approx \frac{f_i(\mathbf{m} + \delta_j \Delta m) - f_i(\mathbf{m})}{\Delta m}$$

Iterative solution of non-linear problems

Starting with a model $\mathbf{m}_0$, we iteratively ($n = 0, 1, \dots$) improve the model by

$$\mathbf{m}^{n+1} = \mathbf{m}^n + \Delta \mathbf{m}^n$$

by solving the linear sub-problem

$$\mathbf{S}^n \Delta \mathbf{m}^n = \Delta \mathbf{d}^n = \mathbf{d} - \mathbf{f}(\mathbf{m}^n)$$

The objective function

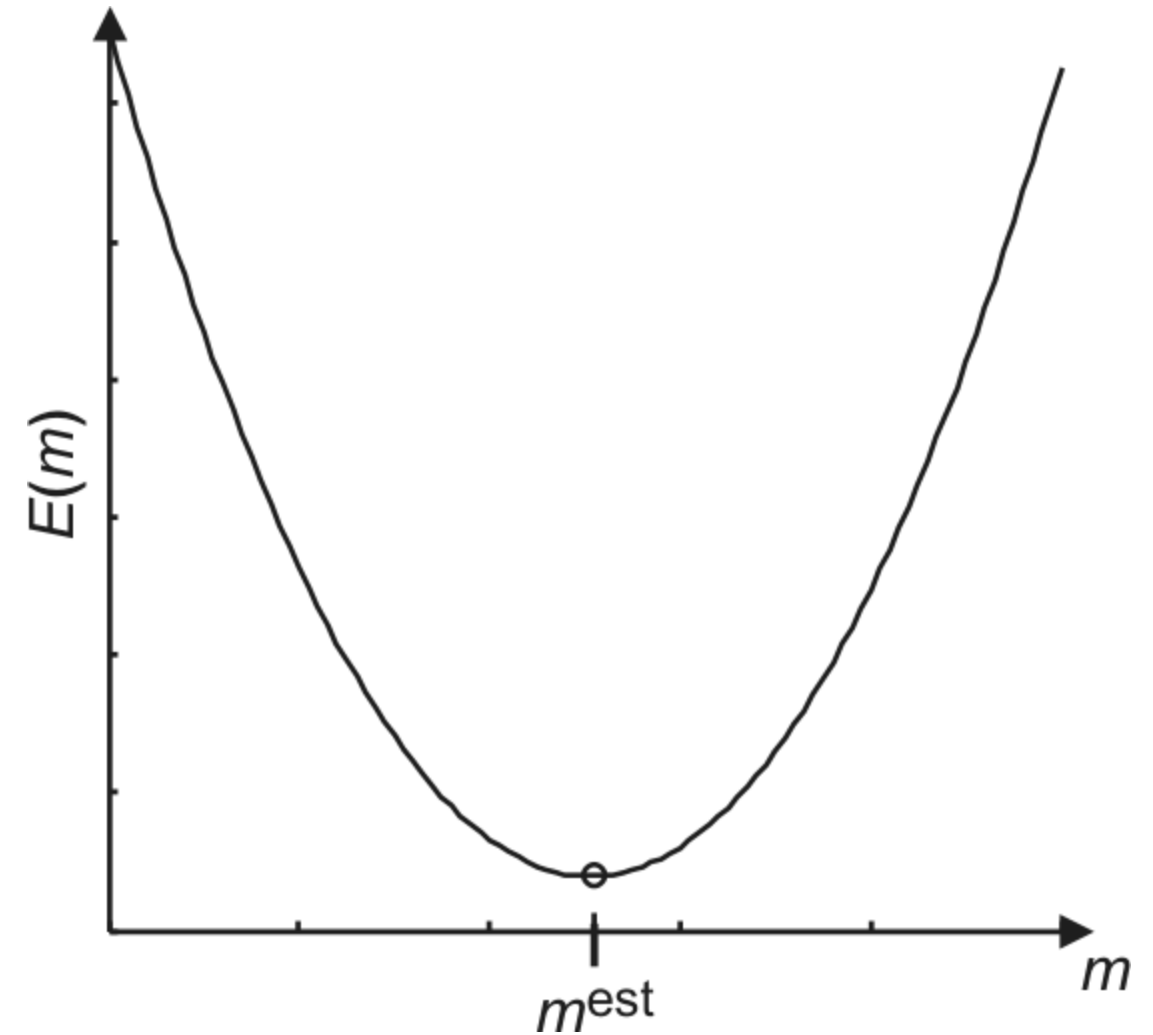
$$\Phi_d = \|\mathbf{d} - \mathbf{f}(\mathbf{m})\|^2 = (\mathbf{d} - \mathbf{f}(\mathbf{m}))^T \cdot (\mathbf{d} - \mathbf{f}(\mathbf{m}))$$

Topography of Φ_d determines non-linearity (s. Notebook on VES)

Objective function for linear problems

$$\begin{aligned}\Phi_d &= (\mathbf{d} - \mathbf{G}\mathbf{m})^T (\mathbf{d} - \mathbf{G}\mathbf{m}) = \\ &= (\mathbf{d}^T - \mathbf{m}^T \mathbf{G}^T)(\mathbf{d} - \mathbf{G}\mathbf{m}) = \\ &= \mathbf{d}^T \mathbf{d} + \mathbf{m}^T \mathbf{G}^T \mathbf{G} \mathbf{m} - 2\mathbf{d}^T \mathbf{G} \mathbf{m}\end{aligned}$$

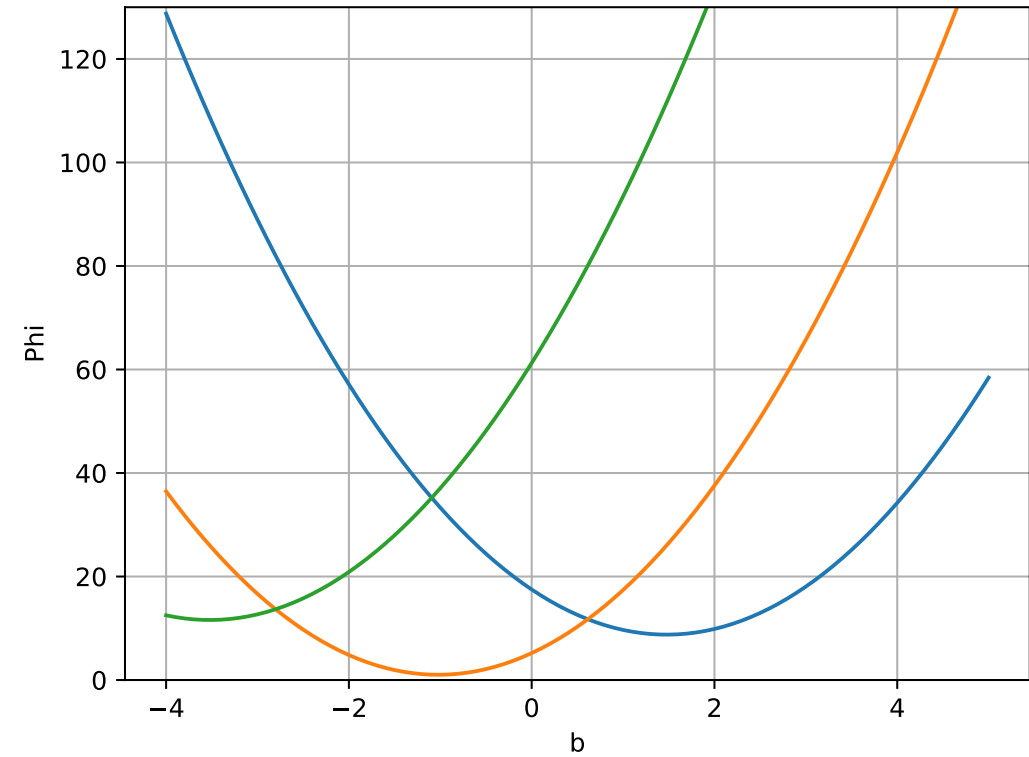
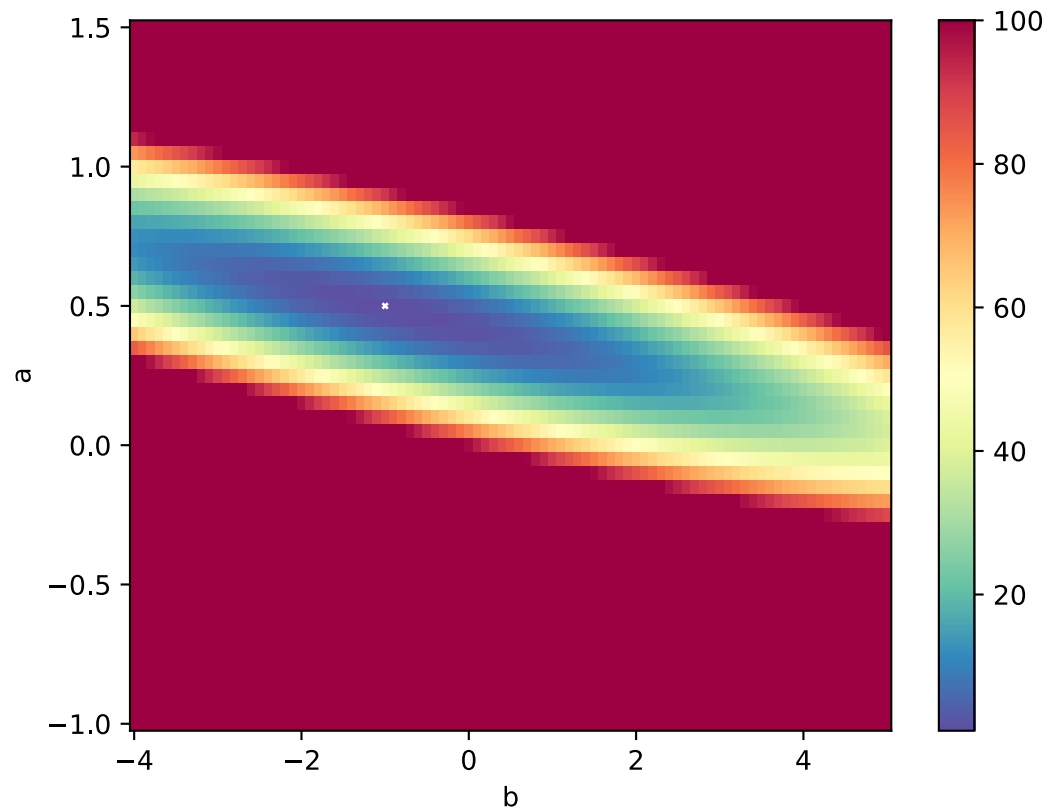
$\Rightarrow$ a parabola for all m_i



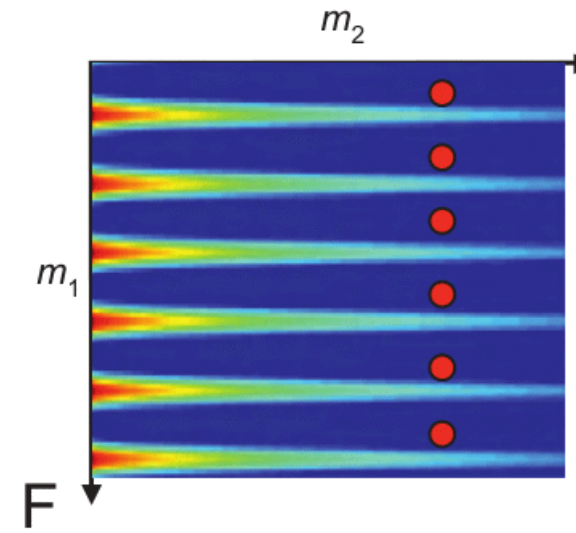
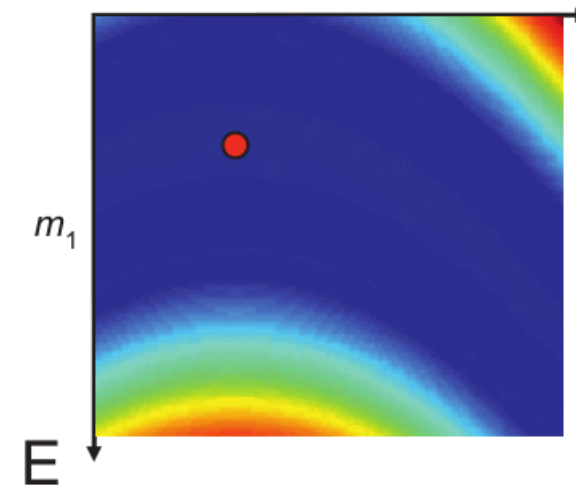
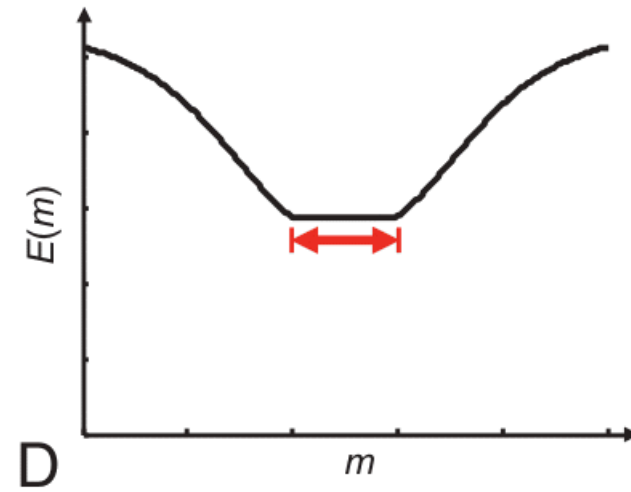
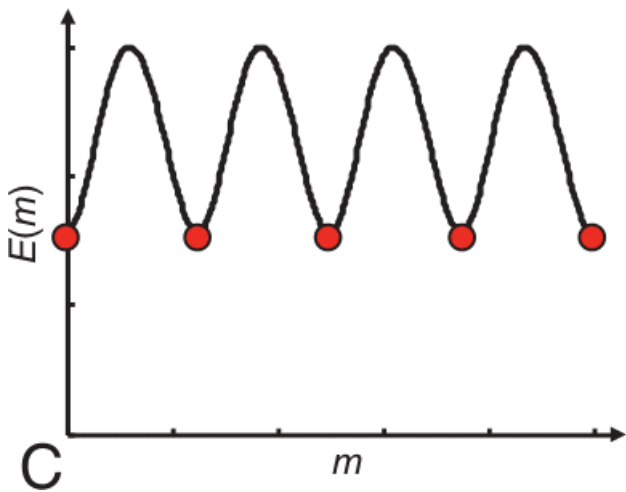
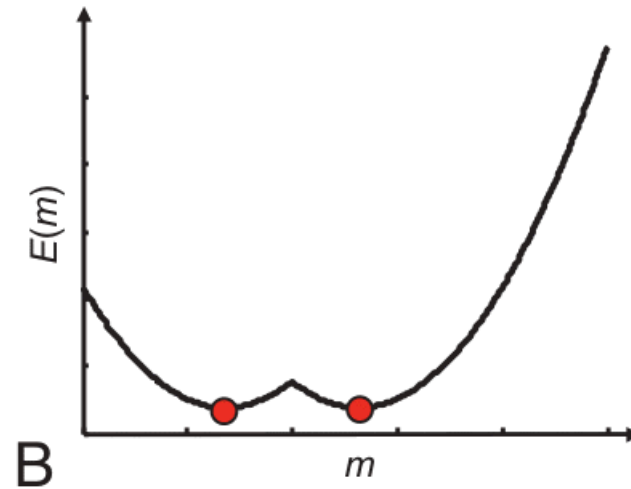
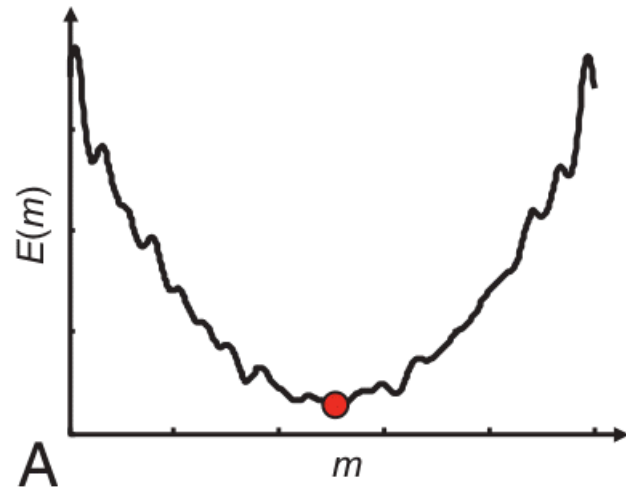
Menke (2012)

Example: linear regression

$$y = ax + b \quad (a=0.5, b=-1)$$

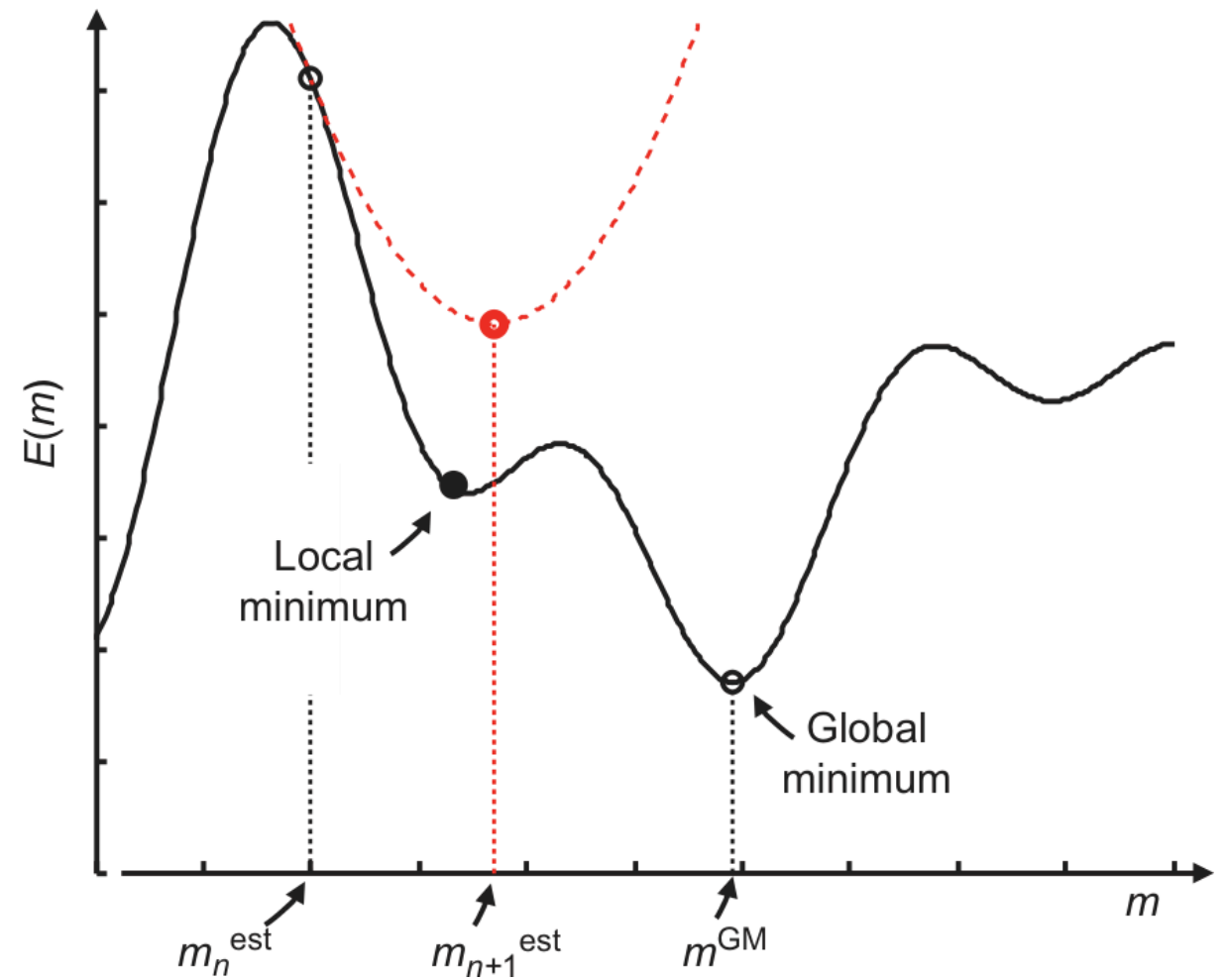
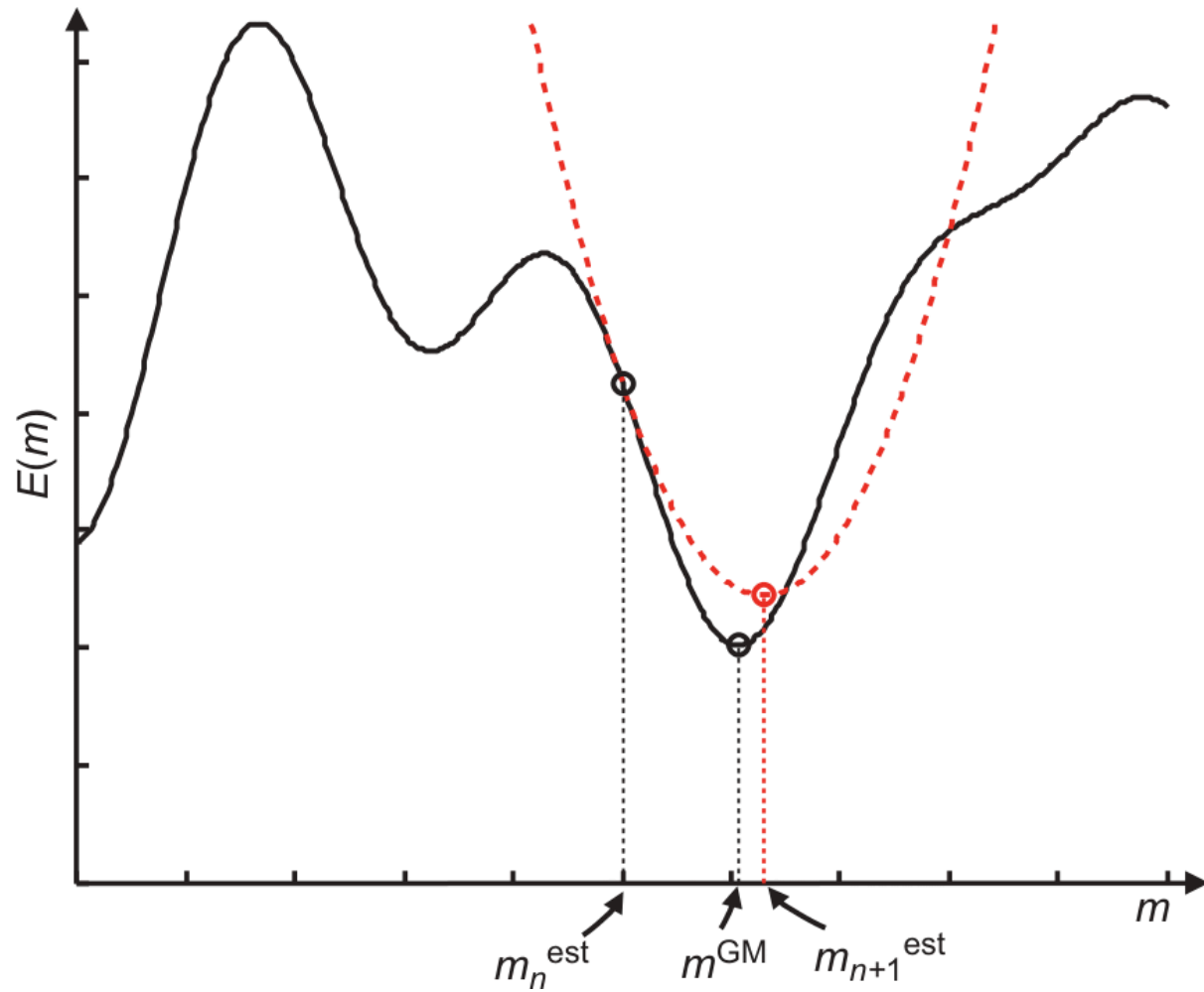


Misfit for non-linear problems (Menke, 2012)



A single minimum
B multiple minima
C periodics
D broad range
E flat valley
F separated valleys

Linearization (Menke, 2012)



Line search for strong non-linearity

The Taylor approximation might change along $\Delta \mathbf{m}$, so we optimize the step length

$$\mathbf{m}^{n+1} = \mathbf{m}^n + \tau^n \Delta \mathbf{m}^n$$

Newton's method

Taylor series expansion for the objective function:

$$\begin{aligned}\Phi_d(\mathbf{m}) = & \Phi_d(\mathbf{m}^0) + \sum_i^M b_i(m_i - m_i^0) + \\ & + \frac{1}{2} \sum_i^M \sum_j^M B_{ij}(m_i - m_i^0)(m_j - m_j^0)\end{aligned}$$

with **gradient vector** $b_i = \frac{\partial \Phi_d}{\partial m_i}$ and **Hessian matrix** $B_{ij} = \frac{\partial^2 \Phi_d}{\partial m_i \partial m_j}$

Newton's method

$$\dots \sum_i^M b_i(m_i - m_i^0) + \frac{1}{2} \sum_i^M \sum_j^M B_{ij}(m_i - m_i^0)(m_j - m_j^0)$$

Minimum: derivative zero

$$\frac{\partial \Phi}{\partial m_k} = 0 = b_k + \sum_j B_{kj}(m_k - m_k^0) \Rightarrow \mathbf{m} - \mathbf{m}^0 = -\mathbf{B}^{-1} \mathbf{b}$$

Newton's method for linear problems

$$\Phi_d = (\mathbf{d} - \mathbf{G}\mathbf{m})^T (\mathbf{d} - \mathbf{G}\mathbf{m})$$

$$b_i = \frac{\partial \Phi_d}{\partial m_i} = -2\mathbf{G}^T (\mathbf{d} - \mathbf{G}\mathbf{m})$$

$$B_{ij} = \frac{\partial^2 \Phi_d}{\partial m_i \partial m_j} = 2\mathbf{G}^T \mathbf{G}$$

$$\Rightarrow \mathbf{m} = -\mathbf{B}^{-1} \mathbf{b} = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T \mathbf{d} \text{ (least-squares solution)}$$

Newton's method for non-linear problems

$$\Phi_d = (\mathbf{d} - \mathbf{f}(\mathbf{m}))^T (\mathbf{d} - \mathbf{f}(\mathbf{m}))$$

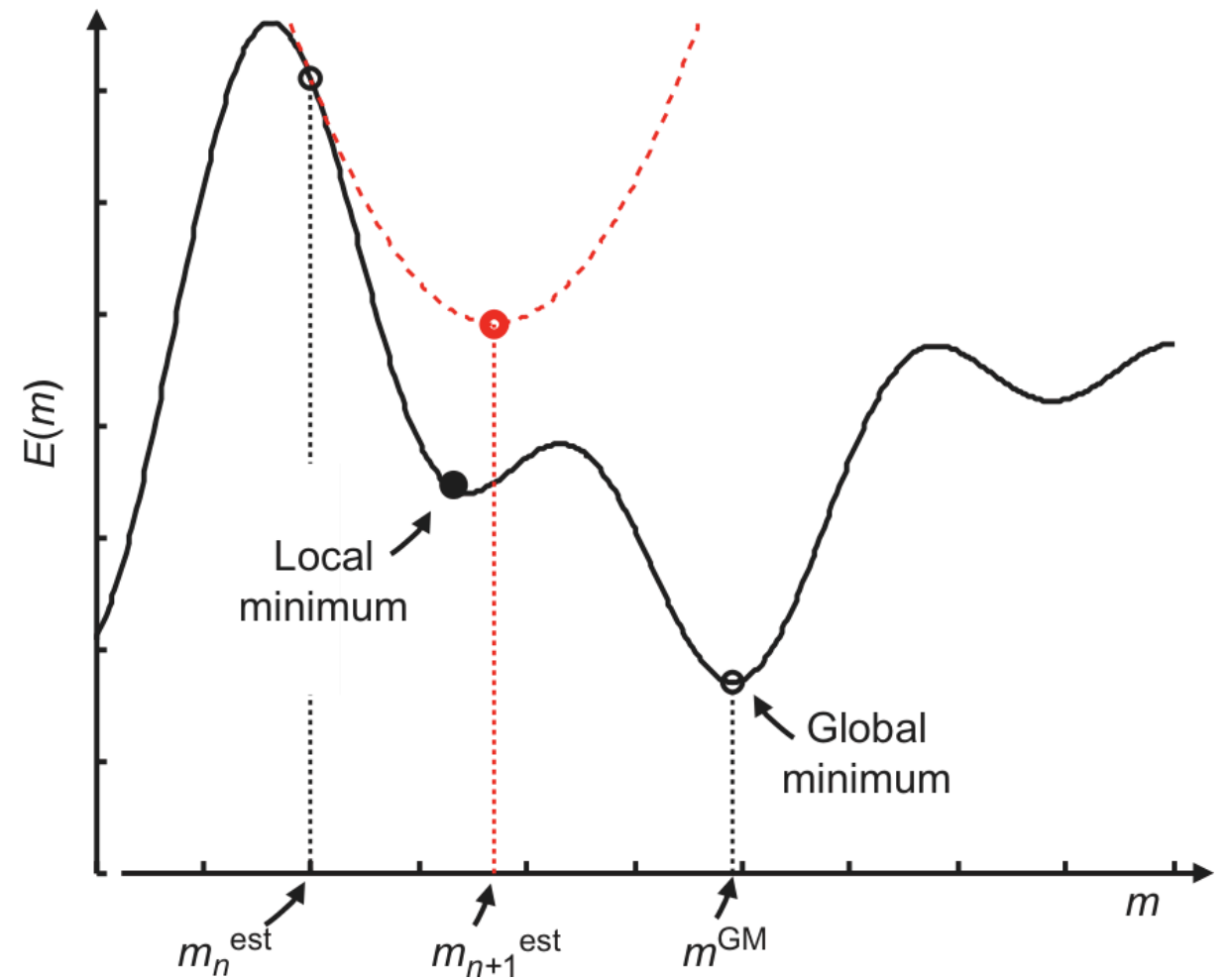
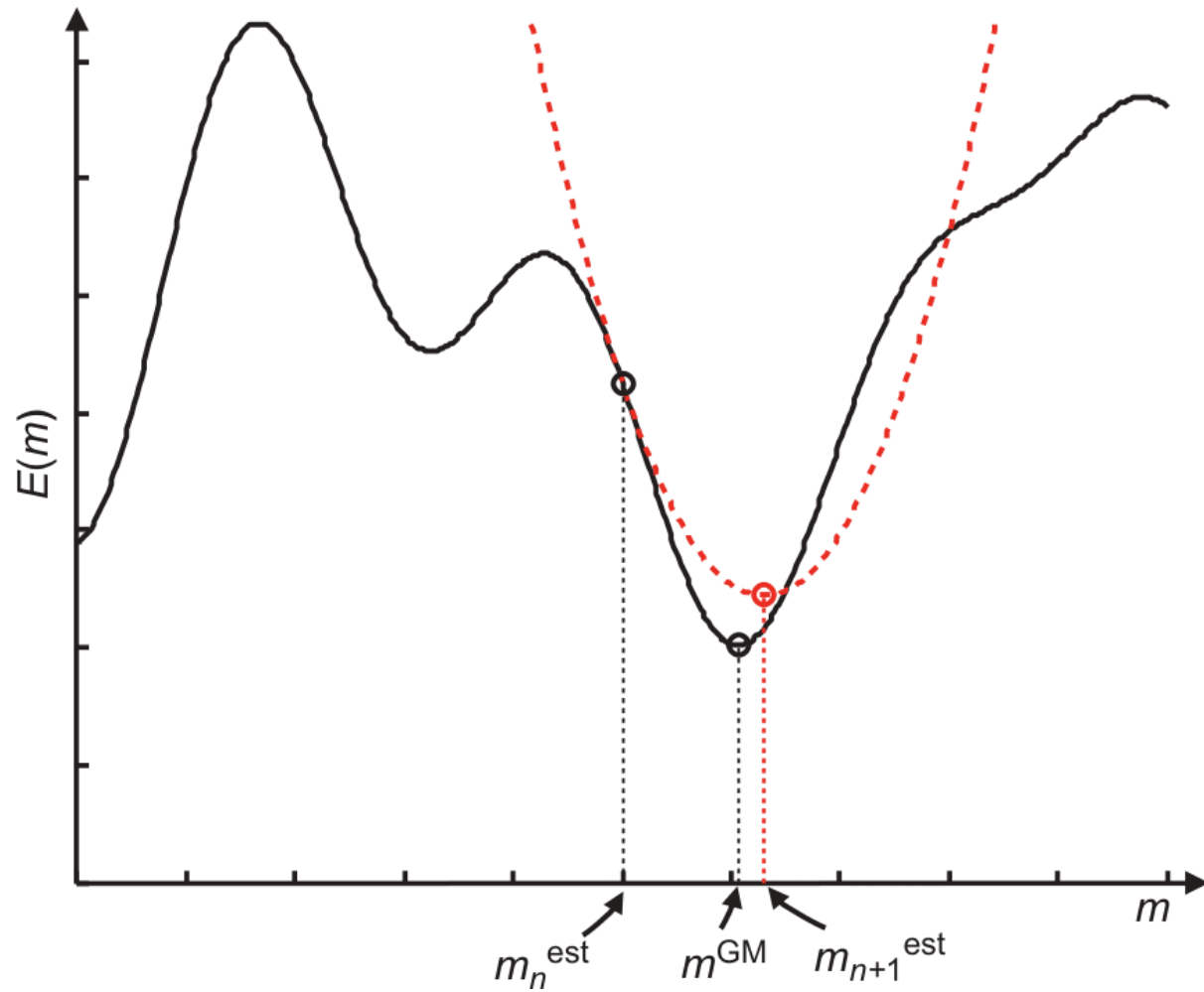
$$b_i = \frac{\partial \Phi_d}{\partial m_i} = -2\mathbf{S}^T (\mathbf{d} - \mathbf{f}(\mathbf{m})) \quad \text{with} \quad S_{ij} = \frac{\partial f_i(\mathbf{m}^0)}{\partial m_j}$$

$$B_{ij} = \frac{\partial^2 \Phi_d}{\partial m_i \partial m_j} = \frac{\partial b_i}{\partial m_j} \approx 2\mathbf{S}^T \mathbf{S}$$

$$\Rightarrow \mathbf{m} - \mathbf{m}^0 = -\mathbf{B}^{-1} \mathbf{b} = (\mathbf{S}^T \mathbf{S})^{-1} \mathbf{S}^T (\mathbf{d} - \mathbf{f}(\mathbf{m}))$$

$\Rightarrow$ least-squares solution for $\Delta \mathbf{m} / \Delta \mathbf{d}$

Linearization (Menke, 2012)



Gauss-Newton minimization

1. Choose starting model $\mathbf{m}^0$ and set $n=0$
2. Compute model response $\mathbf{f}(\mathbf{m}^n)$
3. Compute sensitivity matrix $\mathbf{S}^n$
4. Solve linearized subproblem $\mathbf{S}^n \Delta \mathbf{m}^n = \Delta \mathbf{d} = (\mathbf{d} - \mathbf{f}(\mathbf{m}^n))$
5. Update model by $\mathbf{m}^{n+1} = \mathbf{m}^n + \Delta \mathbf{m}^n$
6. If convergence quit, otherwise $n \leftarrow n + 1$ & proceed with 2.

Line search for strong non-linearity

The Taylor approximation might change along $\Delta \mathbf{m}$, so we optimize the step length τ by “searching along the line”

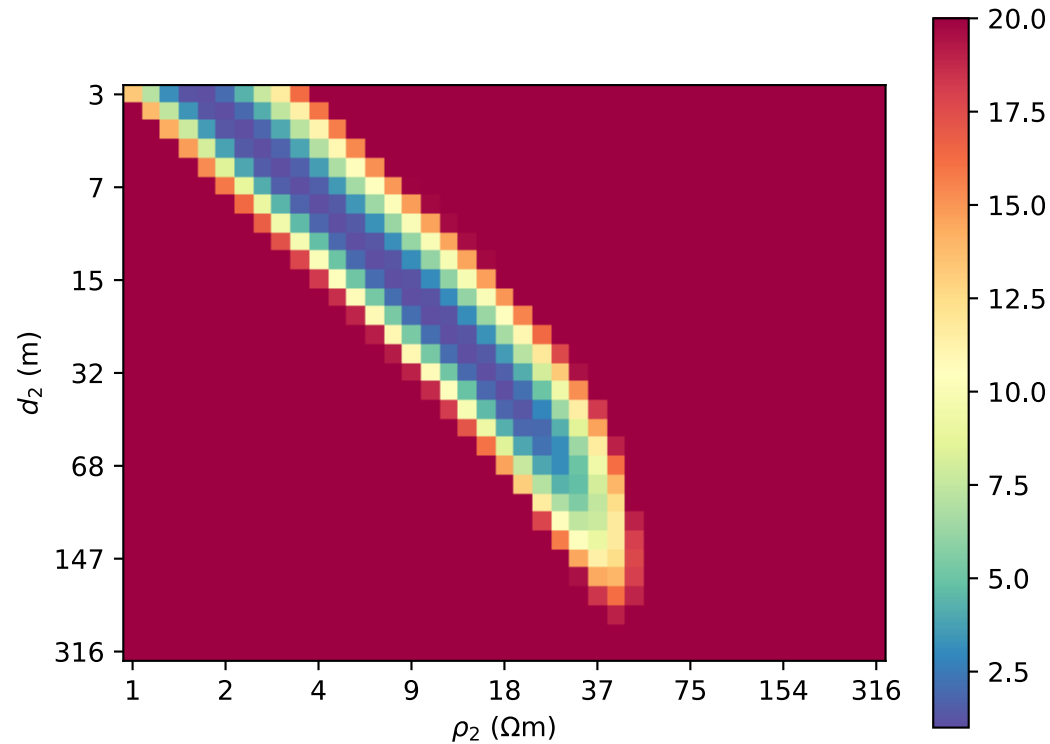
$$\mathbf{m}^{n+1} = \mathbf{m}^n + \tau^n \Delta \mathbf{m}^n$$

- exact line search: testing $\mathbf{f}(\mathbf{m} + \tau \Delta \mathbf{m})$ for many τ
- interpolation of $\mathbf{f}(\mathbf{m}^n + \tau \Delta \mathbf{m})$ between $\mathbf{f}(\mathbf{m}^n)$ & $\mathbf{f}(\mathbf{m}^n + \Delta \mathbf{m})$
- fit parabola through points $\mathbf{f}(\mathbf{m}^n)$, $\mathbf{f}(\mathbf{m}^n + \Delta \mathbf{m}/2)$, $\mathbf{f}(\mathbf{m}^n + \Delta \mathbf{m})$

Gauss-Newton minimization

1. Choose starting model $\mathbf{m}^0$ and set $n=0$
2. Compute model response $\mathbf{f}(\mathbf{m}^n)$
3. Compute sensitivity matrix $\mathbf{S}^n$
4. Solve linearized subproblem $\mathbf{S}^n \Delta \mathbf{m}^n = \Delta \mathbf{d} = (\mathbf{d} - \mathbf{f}(\mathbf{m}^n))$
5. Optimize line search parameter τ^n
6. Update model by $\mathbf{m}^{n+1} = \mathbf{m}^n + \tau^n \Delta \mathbf{m}^n$
7. If convergence quit, otherwise $n \leftarrow n + 1$ & proceed with 2.

Model transformation



Objective function for VES

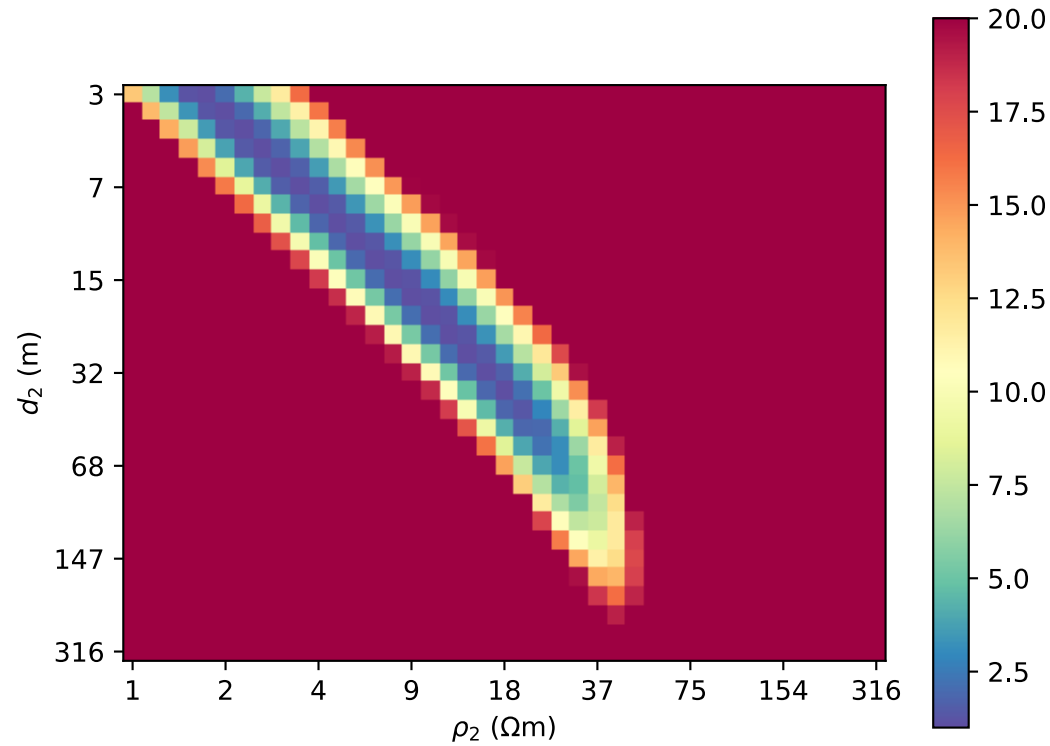
To keep the parameters positive, we often invert for the logarithms.

If we invert for $\hat{m}$ instead of m , we use the chain rule

$$\frac{\partial f}{\partial \hat{m}} = \frac{\partial f}{\partial m} \cdot \frac{\partial m}{\partial \hat{m}} = \frac{\partial f}{\partial m} / \frac{\partial \hat{m}}{\partial m}$$

$$\text{E.g. } \partial \log \rho / \partial \rho = 1/\rho$$

Data transformation



Objective function for VES

Often, measured data show a wide range so that we use the logarithm

If we invert $\hat{d}$ instead of d , we use the chain rule

$$\frac{\partial \hat{f}}{\partial m} = \frac{\partial f}{\partial m} \cdot \frac{\partial \hat{f}}{\partial f}$$

$$\text{E.g. } \partial \log \rho^a / \partial \rho^a = 1 / \rho_a$$

Combined model and data transformation

$$\text{Sensitivity } S_{ij} = \frac{\partial \rho_i^a}{\partial \rho_j}$$

$$\text{Data } d_i = \log \rho_i^a, \text{ model parameter } m_i = \log \rho_j$$

⇒ Jacobian matrix

$$J_{ij} = \frac{\partial \log \rho_i^a}{\partial \log \rho_j} = \frac{\partial \rho_i^a}{\partial \rho_j} \cdot \frac{\rho_j}{\rho_i^a}$$

Regularization

$$\Phi = (\mathbf{d} - \mathbf{f}(\mathbf{m}))^T (\mathbf{d} - \mathbf{f}(\mathbf{m})) + \lambda (\mathbf{c} - \mathbf{C}\mathbf{m})^T (\mathbf{c} - \mathbf{C}\mathbf{m})$$

$$b_i = \frac{\partial \Phi}{\partial m_i} = -2\mathbf{S}^T (\mathbf{d} - \mathbf{f}(\mathbf{m})) - 2\lambda \mathbf{C}^T (\mathbf{c} - \mathbf{C}\mathbf{m})$$

$$B_{ij} = \frac{\partial^2 \Phi}{\partial m_i \partial m_j} = \frac{\partial b_i}{\partial m_j} \approx 2\mathbf{S}^T \mathbf{S} + 2\lambda \mathbf{C}^T \mathbf{C}$$

$$\Rightarrow (\mathbf{S}^T \mathbf{S} + \lambda \mathbf{C}^T \mathbf{C}) \Delta \mathbf{m} = \mathbf{S}^T (\mathbf{d} - \mathbf{f}(\mathbf{m}^n)) + \lambda \mathbf{C}^T (\mathbf{c} - \mathbf{C}\mathbf{m})$$

Summary

- most geophysical problems are non-linear
- linearization around given model
- iterative improvement of the model
- regularization remains (would)